Binder Diver: Core Logic Architecture Specification

Purpose: This document outlines the programmatic logic engine for routing Magic: The Gathering singles to their most likely physical store location (showcase, binder, or bulk box) based on live API parameters. Pass this directly to your coding agent to scaffold the decision-tree logic.

1. Core Inputs & Variables

The algorithm requires the following attributes sourced from the card data payload (e.g., Scryfall API):

• **market_price (float):** Current USD value (e.g., usd or usd_foil).

• **set_type (string):** Identifies core expansions vs. supplemental sets (e.g., 'expansion', 'commander', 'masters').

• **released_at (date):** Used to calculate set age relative to current date.

• **rarity (string):** common, uncommon, rare, mythic.

• **legalities (object):** Format legality arrays (specifically checking 'standard' and 'commander').

2. Decision-Tree Architecture (The Sieve)

Implement a cascading conditional layout to evaluate probability tiers before custom store profiles are implemented. The logic flows from highest-value protection down to lowest-value bulk.

```
def route_card_probability(card):
    # Tier 1: The Display Case (Premium)
    if card.market_price >= 20.00:
        return {"primary": "Display Case", "confidence": 0.90, "secondary": "
$10+ Premium Binder"}

    # Tier 2: The Mid-High Value Color Binders
    elif 5.00 <= card.market_price < 20.00:
        if card.set_type == "commander" or card.legalities.get("standard") ==
 "not_legal":
            return {"primary": "Commander Binder / Box", "confidence": 0.75,
"secondary": "$10+ Premium Binder"}
        return {"primary": "$10+ Premium Binder (Sorted by Color)", "confiden
ce": 0.70, "secondary": "Set-Specific Binder"}

    # Tier 3: The Mid-Tier Floor Mix
    elif 1.00 <= card.market_price < 5.00:
        # Check set age (3 years threshold = 1095 days)
        is_recent_set = (datetime.now() - card.released_at).days <= 1095
```

```
        if card.set_type == "commander":
            return {"primary": "Commander Box / Binder", "confidence": 0.80,
"secondary": "Set-Specific Binder"}
        elif is_recent_set:
            return {"primary": f"Recent Set Binder ({card.set_name})", "confi
dence": 0.70, "secondary": "Sorted Color Boxes"}
        else:
            return {"primary": "Sorted Boxes (by Color)", "confidence": 0.60,
 "secondary": "A-Z Bulk Boxes"}

    # Tier 4: Bulk Pit
    else:
        is_recent_set = (datetime.now() - card.released_at).days <= 1095
        if is_recent_set:
            return {"primary": f"Recent Bulk Row Box ({card.set_name})", "con
fidence": 0.65, "secondary": "General A-Z Bulk Box"}
        return {"primary": "General A-Z Bulk Box", "confidence": 0.80, "secon
dary": "Boxes Sorted by Color"}
```

3. Future Feature Expansion Strategy: Store Layout Profiles

To bypass the fallback matrix later, the database can store a simple configuration object per user/LGS profile. Your agent should structure data schemas to accommodate these adjustments:

Configuration Toggle	Data Type	Impact on Matrix Logic
uses_set_binders	Boolean	If False, completely bypass Recent Set Binder routes and redirect directly to Color Boxes.
has_commander_section	Boolean	If False, routes all supplemental format cards back into standard price tiers.
showcase_threshold	Float	Overrides the hardcoded $20.00 limits for the Display Case bucket.